English translation prepared with ChatGPT (OpenAI, GPT-6), 26 October 2026, for translation of the Bulgarian text into English and LaTeX formatting. The Bulgarian originals remain the reference; translation may contain errors.

Computer Science, 12 April 2025 Test 1 ——— *The exact grading formula depends on the results. As an approximation, use 2 + the number of points. Time allowed: 3 hours. Good luck!*

Problem 1. An urn contains balls numbered $1, 2, \ldots, 14$. Three balls are drawn successively without replacement. Find the probabilities of:

1. (0.25 points) $A = \{\text{all three numbers are even}\}$;
2. (0.25 points) $B = \{\text{the largest number is } 10\}$;
3. (0.25 points) $C = \{\text{the product is divisible by } 3\}$;
4. (0.25 points) $D = \{\text{the numbers are consecutive, regardless of drawing order}\}$.

(0.5 points) Are A, B independent? What is the probability of C given A?

Problem 2. The joint distribution of X, Y is

$Y \backslash X$	1	2	4
-1	$3/15$	$1/15$	0
0	α	2α	$3/15$
1	$1/15$	0	$1/15$

Find:

1. (0.1 points) α and the marginal distributions;
2. (0.2 points) $\mathbb{E}[X]$ and $\mathbb{E}[Y]$;
3. (0.5 points) $\mathrm{Cov}(X, Y)$ and $\rho_{X,Y}$.[1]

Problem 3. Let $X_1, \ldots, X_n$ be iid and $S_n = X_1 + \cdots + X_n$.

1. (0.5 points) Find $\mathbb{E}[X_i \mid S_n]$ for $1 \le i \le n$.
2. (0.25 points) For $X_1 \sim \mathrm{Geo}(p)$, find $g_{S_n}(s)$.
3. (0.25 points) Calculate $\mathbb{P}(X_1 > X_2 + X_3)$ and $\mathbb{P}(X_1 > X_2 + X_3 \mid S_3 = 15)$.

Problem 4. All n seats at a local theatre performance celebrating Culture Day are reserved. Spectators enter one at a time and should sit in their reserved seats. The first spectator is absent-minded and chooses uniformly among all n seats. Each subsequent spectator takes their own seat if available; otherwise they choose a random *free* seat.

1. (0.25 points) Find separately the probabilities that the first, second and third spectators take their own seats.
2. (0.25 points) Vasil is delayed by an examination at FMI and arrives last. What is the probability that he takes his own seat?
3. (0.5 points) Find the probability that spectator k takes their own seat.

[1] If using a calculator, give answers to accuracy 0.0001.